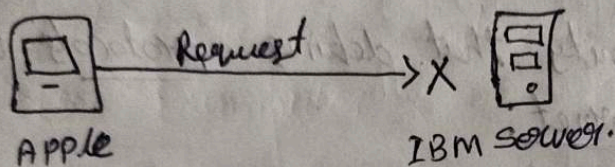


Day 3 - How the TCP/IP model Actually works

⇒ A protocol is set of rules defining how data should be communicated between devices over a network.

- The languages that computers use to communicate.

⇒ Since the early days companies from different companies couldn't communicate with each other.



⇒ A standard is an ~~agreed~~ agreed-upon specification that describes how a protocol or technology ^{should} work.

- with vendor-neutral standards, devices of all types can communicate with each other.

History

⇒ Early 1960,

→ The US department of Defense's ARPA (Advanced Research Projects Agency),

→ which came online in 1969 to connect mainframes at universities & labs

→ originally used a protocol called NCP (network control program).

⇒ Vint Cerf & Bob Kahn (working at DARPA) began developing TCP (transmission control program) in 1974

later divided

- TCP
- IP

ARPANET fully switched to TCP/IP on Jan 1, 1983.

Who define the standards?

participate from engineers at many companies

• IEE (Institute of Electrical & Electronics Engineers)

→ Ethernet (802.3)

→ wifi (802.11)

• IETF (Internet Engineering Task force)

• Open community that defines protocols used on the Internet:

→ TCP, IP, UDP, HTTP, DNS, etc.

• Publishes standards in document called RFC (Requests for comments).

Layered models:

• Networks do a lot of diff. jobs to move data from one computer to another.

→ physical transmission of signals.

→ Local delivery on a LAN.

→ Routing traffic between networks.

→ end-to-end conversations, applications, etc.

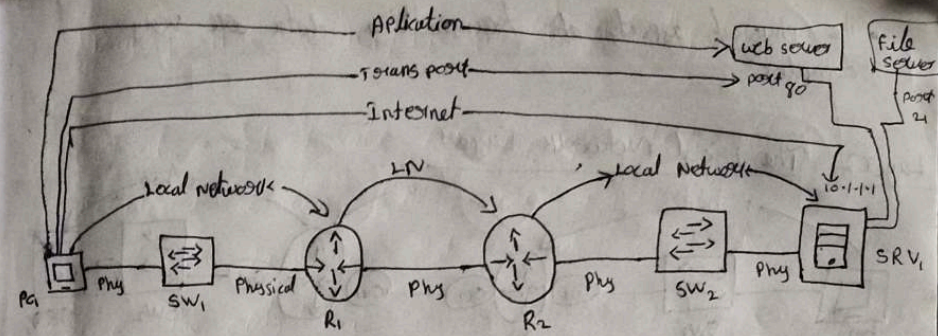
• Five layer model

Application Layer = Telnet, FTP, TFTP

Transport Layer (TCP, UDP) = TCP, UDP

Internet Layer (IPv4, IPv6) = Internet protocol & ICMP

Link Layer (Ethernet, wifi) = Local Network protocol.



Application Layer

• protocols for communication between application processes; create & interpret the data

Transport Layer

• provides end-to-end communication between application processes using Port numbers

Internet Layer

• provides end-to-end communication between hosts across network using IP addresses & gateways.

Local network Layer

• provides hop-to-hop delivery within a local network using MAC addresses & switches

Physical Layer

• sends bits as electrical, optical, or radio signals over the physical medium

Layer 1: The Physical layer

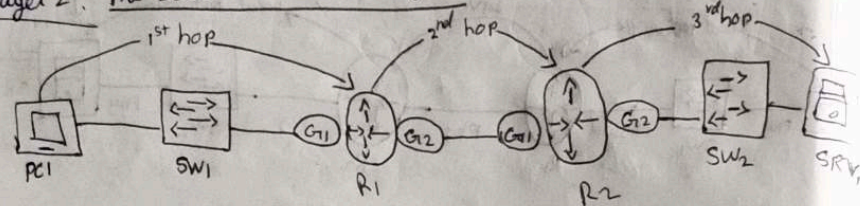
⇒ Sends & Receives bits as electrical, optical, or radio signals over the medium.

⇒ Define things like cables, connectors, signal levels, & link speed

⇒ Ex: Copper UTP cables, fiber-optic cables, wifi radios & antennas, network interface cards (NIC)

- The physical aspects of transmitting data are very complex

Layer 2: The Local Network Layer



- Layer 2 provides hop-to-hop delivery of messages on local networks

→ A hop is one step along the path between two devices

- from one router or host, to the next router or host in the path.

→ switches don't count; a switch just extends the local network, allowing multiple devices to connect

- uses MAC address to identify interfaces

→ PC1 sends the message to the MAC address of R1's G11 interface (NIC)

→ R1 " " " " R2's G12 interface (NIC)

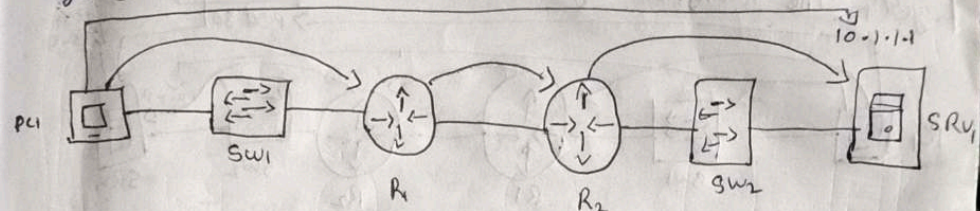
→ R2 " " " " SRV1 interface (NIC)

- protocols at this layer include:

- Ethernet (IEEE 802.3)

- Wi-Fi (IEEE 802.11)

Layer 3: The Internet Layer



- The L3 provides end-to-end delivery between hosts across multiple networks

Internet = Internetwork (between networks)

- uses IP addresses to identify hosts in the network

when PC1 sends a message to SRV1, it addresses the message to SRV1's IP address.

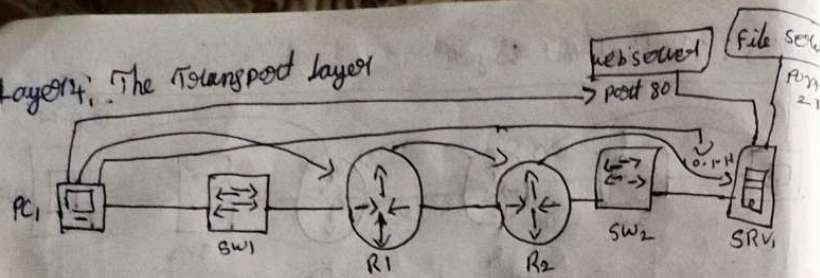
- Routers operate mainly at this layer, using the message's destination IP address to forward the message toward its final destination host.

- protocols at this layer

- IP (IPv4, IPv6)

- ICMP (Internet control message protocol)

Layer 4: The Transport Layer



- L4 provides end-to-end communication between application processes.

→ Also called "process-to-process" or "Service-to-Service."

- Uses Port numbers to identify the processes on each host.

→ When the web client on PC1 wants to send a request to the web server running on SRV1, it addresses the message to port 80.

- Runs mainly on the communicating hosts (PC1 & SRV1); routers normally operate based on IP (L3), not on T-L information.

- Protocols at this layer include:

→ UDP (User Datagram Protocol): simple & efficient
 → TCP (Transmission Control Protocol): more robust features beyond basic message addressing

Layer 5: The Application Layer

- Where network communications meet applications.
- Usually called L7

- Defines how applications process format, send, & interpret data

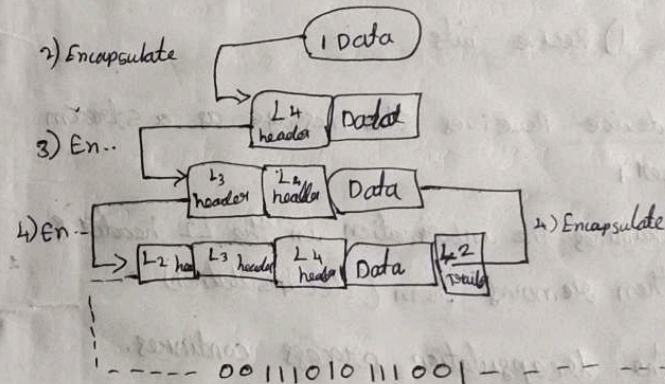
- Protocols at this layer define message formats & rules for specific tasks, such as:

→ Browsing web pages (HTTP/HTTPS)
 → Transferring files (FTP, TFTP)
 → Send/receive emails (SMTP, POP3, IMAP)

- Routers & SW don't care about Application-layer details

→ They just move messages across the network
 → Only the communicating hosts interpret the data

Encapsulation & decapsulation



- The Application layer prepares the data to be sent over the network

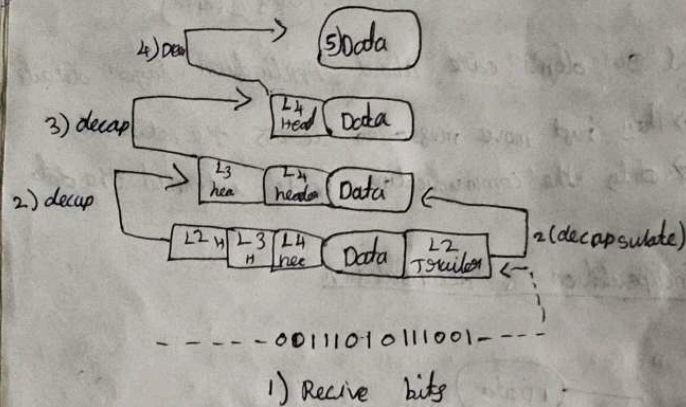
- As the message moves down the stack, each layer encapsulates the data with a header including the info. need for that layer

→ Source & destination addresses (port numbers, IP address, MAC addresses)

→ Layer 2 also adds a trailer that the receiving device uses to check for transmission errors

- The Physical layer transmits the bits as signal over the physical medium.

→ The L2 header is transmitted first, and the L2 transmitted ~~bits~~ that.

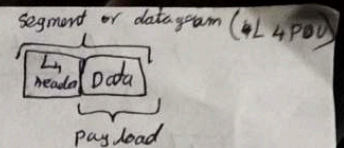


- The Receiving device receives the message as a stream of bits at Layer 1.
 - The device examines the information in the L2 header & trailer & then removes them (decapsulation).
- The decapsulation process continues.

Protocols & data units

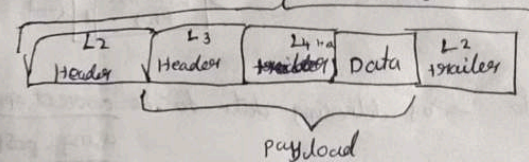
- At each stage in the en/deca process, there is a name given to the message:
- The combination of data & L₄ header is called segment (TCP) or datagram (UDP)

→ TCP creates segments
 → UDP creates datagrams



- The combination of segment/datagram and L3 header is called packet.

- The combination of a packet & L2 header/trailer is called a frame.
- This is actually sent over the wire

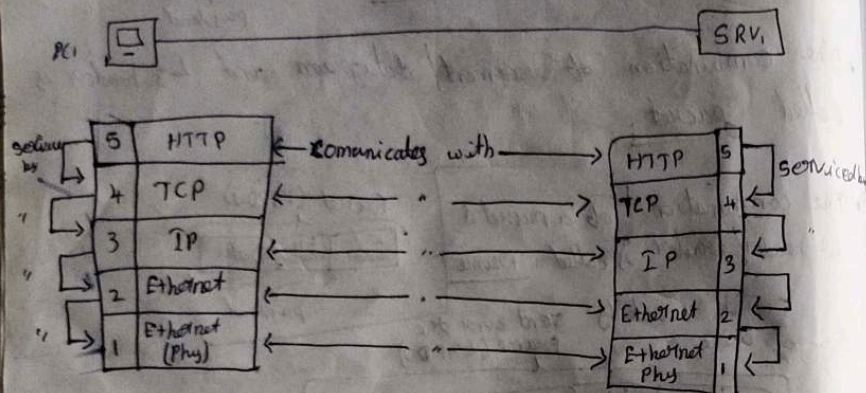


- Alternative name: protocol data unit (PDU)
- A segment or datagram is Layer 4 PDU (L4 PDU)
- A packet is Layer 3 PDU (L3 PDU)
- A frame is Layer 2 PDU (L2 PDU)

- The contents of each PDU (everything encapsulated by that layer's header/trailer) are called the payload.

- A segment or datagram's payload is Application data
- A packet's payload is segment or datagram
- A frame's payload is packet.

Adjacent-layer interaction



- L4 provides service to L5 by delivering data to the correct application using port number
- L3 " " L4 " delivering seg/datagram to correct application destination host using IP address
- L2 " " L3 " packets to the next hop using MAC Addresses
- L1 " " L2 by sending & receiving frames as electrical, optical, or radio signals
- Each Layer communicates with same layer on other devices (same layer interaction)
 - The Application layer on one host sends data to the Application layer on the other host

→ A segment / datagram is addressed to the Layer 4 port number of the correct application on the destination host

→ A packet is addressed to the L3 IP address of the destination host.

→ A frame is addressed to the L2 MAC address of next hop

→ signals sent out of physical ports are received by physical port on the connected device.

Review

Encapsulation & deapsulation processes.

* In Encap. the sending host adds headers & trailer, to the data to prepare it for transmission over the physical medium.

* In decap. Receiving host removes the headers and trailer layer by layer until gets to the data inside.